Chapter 6 : Query Languages

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Keyword-based Querying

- Queries are *combinations of words*.
- The document collection is searched for documents that *contain these words*.
- Word queries are *intuitive*, *easy* to express and provide *fast ranking*.
- The *concept of word* must be defined:
  - A word is a *sequence of letters* terminated by a separator (period, comma, space, etc).
  - Definition of *letter* and *separator is flexible*; e.g., hyphen could be defined as a letter or as a separator.
  - Usually, *common words* (such as “a”, “the”, “of”, …) are ignored.

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Single-word Queries

- A query is a *single word*.
  - Usually used for *searching* in document images.

- *Simplest* form of query.
- What are the possible documents *retrieved as relevant*?
  - All documents that *include this word* are retrieved.

- On what base *documents are ranked*?
  - Documents may be *ranked by the frequency* of the *query word* in the document.
  - Documents containing *more of the query word* are given the *highest priority*.

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Phrase Queries

- A query is a *sequence of words* treated as a *single unit*. Also called "*literal string*" or "*exact phrase*" query.
  - Phrase is usually surrounded by *quotation marks*.
  - All documents that include this *phrase are retrieved*.
- Usually, separators (commas, colons, ...) & common words ("a", "the", "of", "for"…) in the *phrase are ignored*.
- In effect, this query is for a *set of words* that must *appear in sequence*.
  - Allows users to *specify a context* and thus *gain precision*.
  - Ex.: "Information Processing for Document Retrieval".
- What are the possible *documents retrieved* as *relevant*?
  - All documents that include *phrase query* are retrieved.
- On what base *documents are ranked*?

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Multiple-word Queries

- A query is a *set of words* (or phrases).
  - Ex.: What is the result for the query "*Data Mining and Intelligent Database Design*"?

- What are the possible documents *retrieved as relevant*?
  - *Two options*: A document is retrieved if it includes:
    - Any of the query words, or
    - each of the query words.

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Multiple-word Queries

- On what bases documents be *ranked to list* according to *best matching principle*?
  - Documents are ranked by the number of *query words* they contain.
  - A document containing n query words is ranked higher than a document containing *m < n query words*.

  - Documents are ranked in *decreasing order*:
    - Those containing all the *query words are ranked* at the top, *only one query word at bottom*.

  - *Frequency counts* may be used to break ties among documents that contain the *same query words*.

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Boolean Queries

- Queries are formulated based *on concepts from logic*: *AND*, *OR*, *NOT*.
  - It describes the *information needed* by relating multiple words with *Boolean operators*.
- ***Semantics***: For each query word w a corresponding set D_w is constructed that includes the *documents that contain w*.
- The *Boolean expression* is then interpreted as an expression on the corresponding document sets with *corresponding set operators*:
  - ***AND***: Finds only documents containing *all of the specified words* or *phrases*.
  - ***OR***: Finds documents containing *at least one of the specified words* or *phrases*.
  - ***NOT***: Excludes documents containing the *specified word* or *phrase*.

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Examples: Boolean Queries

- 1.Computer OR server
  - Finds documents containing *either computer*, *server* or *both*.
- 2. (computer OR server) NOT mainframe
  - Select all documents that discuss *computers or servers*, do not select any documents that *discuss mainframes*.
- 3. Computer NOT (server OR mainframe)
  - Select *all documents that discuss computers*, and do not discuss either servers or mainframes.
- 4. Computer OR server NOT mainframe
  - Select all documents that discuss *computers*, or documents that discuss *servers* but *do not discuss mainframes*.

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Weighted Queries

- Each of the words is assigned a *different weight*, expressing the *relative importance* of the word within the query.
- A query is then a *set of word-weight pairs*:

$$(q_1, w_1), (q_2, w_2), \ldots, (q_n, w_n).$$

- The *ranking* of a document is the *sum of the weights* for the query words that it satisfies.
- ***Example***: given Query: (A,0.8,), (B,0.9), (C,0.3); and
- ***Document 1***: (A, B, D) and ***Document 2***: (A, C, D) which document *ranked first* ?
  - Score of Document 1: 0.8+0.9 = 1.7
  - Score of Document 2: 0.8+0.3 = 1.1
  - Each document includes *two words from the query*, but Document1 is ranked higher because it includes *more important words*.

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Penalizing Documents

- When interpreting *queries*,
  - The Boolean model does not "penalize" documents with *extra* (non-requested) *keywords*.
  - Some models demote documents that include *keywords* that were *not requested*.
    - The *vector model* with the cosine measure
    - The *probabilistic* Bayesian network model
- Ex.: Assume the vector model with the *cosine measure* and the *simple case* that both documents and queries use *binary values*.
  - Consider the following *two documents* and a *query*:
    - $d_1 = (0,1,0,1,0)$, $d_2 = (0,1,1,1,0)$, $q = (0,1,0,1,0)$
    - $sim(q, d_1) = 1.0$, $sim(q, d_2) = 0.82$
    - d_2 is demoted because it includes an *extra keyword not requested* by q.

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Pattern Queries

- What is Pattern?
  - An expression that defines a *set of objects*. Pattern shows the *internal representation of an object*.
  - What is the *pattern of a word*?

- *Pattern matching*: A word matches a pattern if it is equal to one of the words defined by *the pattern*.

- In other words,
  - The semantics are of *disjunction*: A pattern *P* that defines a *word* $(c_1, c_2, \ldots, c_n)$ is *interpreted* as $c_1 \vee c_2 \vee \ldots \vee c_n$.

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Pattern Queries

- ***Similarity pattern***. Specifies a string and a radius
  - Defines all the words whose *distance from* the string is within the radius.
  - Assume the distance between two strings is measured by the number of *one-character changes* (insertions, deletions, replacements) required to transform one string into the other.
  - The *similarity pattern* (*king*, 2) defines *kin*, *kong*, *knig*, *kings*, *cling*, …
- Useful to compensate for *typing or scanning* (OCR) errors.
  - One of the technique used for pattern matching is *string editing*.

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String Editing

- The problem is given *two sequences of symbols*, $X = x_1\ x_2 \ldots x_n$ and $Y = y_1\ y_2 \ldots y_m$, transform X to Y, based on a sequence of three operations: *Delete*, *Insert* and *Replace*, so that for every operation $COST(C_{ij})$ is incurred.
- The objective of string editing is to identify a *minimum cost sequence* of edit operation that will transform X into Y.
- Example: consider the *sequences*:
  - X = {a a b a b} and Y = {b a b b}
- Identify a *minimum cost sequence* of edit operation that transform X into Y.
- Assume change *costs 2 units*, delete *1 unit* and insert *1 unit*.

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Dynamic programming

- The minimum cost of any *edit sequence that transforms* $x_1\ x_2 \ldots x_i$ into $y_1\ y_2 \ldots y_j$ (for i>0 and j>0) is the minimum of the three costs: *delete*, *replace*, or *insert operations*.
- The following *recurrence equation* is used for COST(i,j).

$$
\text{COST}(i,j) = \begin{cases} 0 & \text{if } i=0,\ j=0 \\ \text{COST}(i-1,0) + D(x_i) & i>0,\ j=0 \\ \text{COST}(0,j-1) + I(y_j) & j>0,\ i=0 \\ \text{COST}'(i,j) & i>0,\ j>0 \end{cases}
$$

$$
\text{where COST}'(i,j) = \min \{ \begin{array}{l} \text{COST}(i-1,j) + D(x_i), \\ \text{COST}(i-1,j-1) + C(x_i,y_j), \\ \text{COST}(i,j-1) + I(y_j) \end{array} \}
$$

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Example

- Transform the *sequences*:
  - Xi = {a a b a b} into Yj = {b a b b}
- With *minimum cost sequence* of edit operation using dynamic programming approach, Assume that *change costs 2 units*, *delete* and *insert* 1 unit.

i \ j	0	1	2	3	4
0	0	1	2	3	4
1	1	2	1	2	3
2	2	3	2	3	4
3	3	2	3	2	3
4	4	3	2	3	4
5	5	4	3	2	3

- The value 3 at (5,4) is the optimal solution
- By tracing back one can determine which operations lead to *optimal solution*.
- Delete x_1, Delete x_2 and Insert y_4 Or,
- Change x_1 to y_1 & Delete x_4.

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Natural language

- Using natural language for querying is *very attractive*.
- ***Example***: Find all the documents that discuss

- " campaign finance reforms, including documents that discuss violations of campaign financing regulations.
- Do not include documents that discuss campaign contributions by the gun and the tobacco industries".

- Natural language queries are converted to a formal language for processing against a *set of documents*.
- Such translation requires *intelligence* and is still a *challenge*.

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Natural language

- Pseudo NL processing: System scans the text and extracts recognized terms and *Boolean connectors*.
- The *grammaticality of the text* is not important.
  - Often used by *search engines*.

- ***Problem***: Recognizing the negation in the *search statement* (“Do not include...”).

- ***Compromise***: Users enter natural language clauses connected with *Boolean operators*.
- In the above example: “*campaign finance reforms” or “violations of campaign financing regulations*" and not “*campaign contributions by the gun and the tobacco industries*”.

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Question & Answer

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Thank You !!!

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